ATTENDX PRO

Smart Attendance Analytics & Alert System

Student Details

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CSE (IoT)

Course Information

OOP Lab

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The Problem

Students lack clarity on their attendance status and exam eligibility until it's too late.

The Solution

Real-time tracking and automated alerts ensure students stay informed and eligible for exams.

SYSTEM ARCHITECTURE

AttendX Pro: Multi-Layered Object-Oriented Design

UI LAYER

User Interface
`(Console)`

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CONTROLLER

AttendXController
`Logic Engine`

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MODEL LAYER

OOP Hierarchy:

- User (Abstract)
  - └ Student
  - └ Faculty

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SERVICES

StorageService
`Data Persistence`

NotificationThread
`Background Alerts`

OOP IMPLEMENTATION

AttendX Pro: Core Object-Oriented Programming Logic

1. Abstraction

Abstract `User` class prevents direct instantiation of generic users.

2. Inheritance

`Student` and `Faculty` extend the base User class.

3. Encapsulation

Data security via private fields and public getter/setter methods.

4. Polymorphism

`getRole()` overridden to return specific user types.

5. Exception Handling

Custom exceptions for robust error management.

6. Collections

`HashMap` utilized for O(1) student lookup.

7. Multithreading

Asynchronous background notification thread for real-time system alerts.

THRESHOLD CALCULATOR

AttendX Pro: Intelligent Attendance Forecasting Logic

THE ATTENDANCE FORMULA

$$(\text{Attended} + x) / (\text{Total} + x) = 75\%$$

The system solves for **x** to determine the exact number of consecutive classes a student must attend to reach the 75% eligibility threshold.

01 **Faculty Login:** Authenticated access to the dashboard.

02 **Mark Attendance:** Instant database updates for sessions.

03 **Student Login:** Personal view of current status.

04 **Eligibility Check:** View target classes needed (x).

PROJECT FINALIZATION

AttendX Pro: Error Handling, Threading & Future Roadmap

ERROR HANDLING

- Duplicate entry exceptions
- Universal input validation
- File I/O recovery logic

BACKGROUND THREAD

- Auto-run every 30 seconds
- Low attendance (<60%) alerts

FUTURE SCOPE

- GUI & Database persistence
- Automated email alerts

CONCLUSION: SYSTEM IMPACT

AttendX Pro successfully simplifies faculty management and provides students with instant, real-time exam eligibility transparency.

THANK YOU!